

Problem Set 5

POLS W4732

Fall 2026

Due November 13 before the TA section

Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

Exercise 1 (Electoral manipulation contract). *A leader offers a local official a wage contingent on whether the leader remains in power, $\pi \in \{0, 1\}$. The official chooses high or low manipulation effort. The leader receives $V\pi - w$ and the official receives $\sqrt{w} - c(e)$; rejection gives the official $\bar{u}$.*

Wages must be nonnegative. When effort is observable, a contract may condition on both effort and the election outcome. When effort is hidden, it may condition only on the election outcome.

Use the parameter tuple assigned to the group; if none is assigned, use A.

Version	p_H	p_L	c_H	$\bar{u}$
A	.60	.50	.10	.20
B	.70	.45	.12	.20
C	.65	.40	.15	.15

In all versions $c_L = 0$.

- 1. An analyst proposes the same flat wage under observable and hidden effort. Identify when it implements low effort and why it generally fails to implement high effort under hidden action.*
- 2. Derive the least-cost contracts implementing low and high effort when effort is observable.*
- 3. Derive the least-cost high-effort contract when effort is hidden. State which constraints bind and when limited liability binds.*
- 4. Characterize the critical value of V at which the leader's preferred implemented effort changes under observable and hidden effort.*

Exercise 2 (General moral hazard). An agent chooses $e \in \{\underline{e}, \bar{e}\}$ after accepting a contract (w_F, w_S) . Success probabilities are $\underline{p} < \bar{p}$. The principal receives $\underline{\pi}$ after failure and $\bar{\pi}$ after success. The agent obtains $v(w) - c(e)$, where $v' > 0$, $v'' < 0$, and has reservation payoff $\underline{u}$.

1. Transform wages into utility units $y_S = v(w_S)$ and $y_F = v(w_F)$. Use binding participation and incentive constraints to solve for (y_S, y_F) when high effort is implemented; then convert with v^{-1} .
2. Write the least-cost low-effort contract and state any limited-liability qualification.
3. Prove, qualify, or refute: (a) high effort always has a higher expected wage; (b) a larger success-probability gap always makes high effort more expensive; (c) if the principal earns less after success than failure net of wages under the high-effort contract, high effort is nevertheless optimal because success is socially valuable.
4. State every assumption needed for your conclusions.